

Idle Sleep & Wake Timing

The calculation behind the **Idle Scanning** diagram: one 5 s heartbeat cycle, repeated.

Interval	MCU	Length	How the sleep is timed
t = 0: heartbeat (and the battery sample every 10 s)	Run 4 MHz	0.5 ms	Stop 2 exit 12.06 µs, LED on, ADC 1.5 ms
Green LED on	Stop 2	10 ms	$\lceil 10 \times 4.096 \rceil = 41$ ticks = 10.01 ms
t = 10 ms: LED off	Run 4 MHz	0.5 ms	pattern step ends; next deadline is the heartbeat
Sleep to the next heartbeat	Stop 2	4 990 ms	$\lceil 4\,990 \times 4.096 \rceil = 20\,440$ ticks = 4 990.2 ms
ST25R3916 antenna check (its own timer)	MCU asleep	every 100 ms	41 µs of field; the MCU sleeps through all 100

How a sleep is timed: LPTIM1 counts the 32.768 kHz crystal ÷ 8 = **4096 Hz**, so one tick is **244 µs**. ticks = $\lceil \text{sleep ms} \times 4096 / 1000 \rceil$: rounded up, so the MCU wakes at most 244 µs late, never early. Stop 2 only for sleeps ≥ 4 ms (shorter ones use Sleep mode). The 16-bit counter wraps every 16 s, so one sleep is at most 15 s; the loop caps it at 5 s.

99.965 %
of the time in Stop 2: 3.5 ms awake in every 10 s

0.4 /s
MCU wake-ups: the LED's on and off edges, every 5 s

12.06 µs
Stop 2 → Run at 4 MHz (13.16 µs max, DS11451)

108 µA
idle average; the reader's checks are 99 µA of it

Awake per 10 s = $4 \times 0.5 \text{ ms} + 1.5 \text{ ms} = \mathbf{3.5 \text{ ms}}$ (0.035 %)
Wake-ups: $0.4/\text{s} \times 0.5 \text{ ms} \times 1.01 \text{ mA} = \mathbf{0.20 \text{ µA}}$
Battery sample: $1.5 \text{ ms} \times 1.21 \text{ mA} \div 10 \text{ s} = \mathbf{0.18 \text{ µA}}$
LED: $2.77 \text{ mA} \times 10 \text{ ms} \div 5 \text{ s} = \mathbf{5.54 \text{ µA}}$
Reader: $9.93 \text{ µC} \times 10/\text{s} = \mathbf{99.3 \text{ µA}}$
Total: $99.3 + 5.5 + 3.3 + 0.2 + 0.2 = \mathbf{108.5 \text{ µA}}$

Card Tap Sleep & Wake Timing

The calculation behind the **One card Tap** diagram: ms after the reader's IRQ, from the scope capture.

Step	ms	MCU	Length	Why it wakes next
Wake-up IRQ (PBI → EXTII)	0 → 1	Run	1.0	Stop 2 exit 12.06 μs; leave wake-up mode, field on
ISO 14443 field guard	1 → 6	Stop 2	5.0	5 ms + 1 tick: $\lceil 5 \times 4.096 \rceil = 21$ ticks
Read: REQA, anticollision, SELECT	6 → 12.2	Run	6.2	SPI exchanges with the reader; field off at the end
Buzz + green LED	12.2 → 62.5	Stop 2	50.3	$\lceil 50.3 \times 4.096 \rceil = 207$ ticks; motor off
Green LED only	62.5 → 262.2	Stop 2	199.7	$\lceil 199.7 \times 4.096 \rceil = 818$ ticks; pattern ends
Empty poll: field on, guard	262.2 → 268.0	Stop 2	5.2	0.5 ms awake to switch the field on; no card in 5.7 ms of field
Re-arm wake-up mode	268.0 → 269	Run	1.0	then Stop 2 until the next heartbeat (≤ 5 s)



96.6 %
of the tap in Stop 2: 260 of 269 ms

9.0 ms
MCU awake in the whole tap, at 1.01 mA

5 wakes
 $5 \times 12.06 \mu\text{s} = 60 \mu\text{s}$ spent waking up

18.0 ms
RF field on (12.2 read + 5.7 empty poll) at 202–228 mA

Awake = $1.0 + 6.2 + 0.3 + 0.5 + 1.0 = 9.0$ ms
MCU charge $\approx 9.0 \text{ ms} \times 1.01 \text{ mA} = 9.1 \mu\text{C}$
Field charge = $2474 + 1302 = 3.78 \text{ mC}$
The motor and LED run while the MCU sleeps: Stop 2 keeps the GPIO levels, so only the pattern's step edges wake it. The radio, not the processor, sets the cost.

Times from the scope capture TEK00006 (1 Ω shunt). The firmware's buzz step is 60 ms (APP_FB_ACCEPT_VIB_MS); the scope saw 50 ms of motor current. Field guard: APP_NFC_FIELD_GUARD_MS = 5 ms + 1 tick. One LPTIM tick = 244 μs.